

A pragmatic stochastic decision model for supporting goods trans-shipments in a supply chain environment

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ABSTRACT

This paper develops a set of decision rules to assist wholesalers to decide whether it is more cost effective to trans-ship urgent outstanding retailer orders from other wholesalers, very fast but at a higher purchase cost, or to order from their suppliers. By considering the uncertainty in demand, it models the total cost encountered by wholesalers, including purchasing, backordering and holding costs in the inventory replenishment process. Unlike previous models that are complex, highly mathematical and difficult to apply, this model provides a pragmatic and less complex method adoptable by ordinary logistics managers and requires input data that are accessible from the previous transaction records of an organization. The application of the proposed decision rules are illustrated considering different scenarios of wholesaler–supplier combinations.

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1. Introduction

Ever increasing market competition forces wholesalers to be ever cost conscious and responsive to the changing needs of the market. A consequence is that wholesalers maintain a low inventory for the purpose of buffering in order to minimize the possible costs. However, the criticality of losing the sales due to stock outages with consequent possible lost profit and decreased customer satisfaction are no less important. To manage these potentials, wholesalers adopt a mix of urgent order lateral trans-shipments from other wholesalers at a higher cost, while at the same time backordering from their usual suppliers to meet the continuing stochastic retailer demands – this produces flexibility in the inventory system. Rules that assist wholesalers' decision making processes have practical importance for inventory management.

Supply chain management concepts that streamline the flow of goods have been the focus of research for a considerable time. Previous research into wholesaler inventory management has presented complex criteria drawn from diverse information sources that are problematic to adopt in real-world practice. There remains a need for simpler and more readily applicable rules for lateral trans-shipment decisions. In a multi-location setting under a continuous review (R,Q) ordering policy Axsäter [3] considers a decision rule for determining how many units should be trans-shipped, depending on the complete state of the system. The decision is optimal and can be repeatedly used as a heuristic. He puts a significant focus on future cost difference for a certain initial state and the highly mathematical

analysis and probability assumptions may not be easily understood by ordinary managers. The models proposed by Evers [9] and Minner et al. [18] consider the case of lost sales rather than backordering under (R,Q) policies. Olsson [21] has considered an optimal (R,Q) ordering policy under complete pooling but due to the problem complexities the optimal solution is restricted to systems with only two locations. Chiu and Huang [6] considered a system with more than two echelons. However, they then restricted focus to a single location of each echelon. Our study considering a multi-location setting, develops decision rules for reactive lateral trans-shipments of urgent demands that cannot be satisfied from the stock on hand. The decision rules for determining whether it is more cost effective to trans-ship urgent orders or to backorder all outstanding orders from suppliers, the size of trans-shipment, the favorite wholesaler and the favorite supplier. Our approach handles the cost difference issue raised above using an alternative approach mainly based on predicted holding and backorder costs in different time periods. We find that the total cost function against the number of units trans-shipped has a linear relationship and presents a close estimation reflecting the cost variations, in real case, it is not a completely linear relation though. This new approach does not undervalue previous scholarly work, it builds on it by proposing a more pragmatic decision model for supply chain environment in wholesaler system. The model can be applied to a real context with multiple wholesalers, and multiple suppliers with variable lead times. It is less complex in calculations and the data requirement that can be fulfilled with previous transaction records data of the organization enables adoption of this model by an ordinary manager. The proposed approach is validated through the illustration of a practical application of the model in different scenarios.

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The next section reviews the relevant literature, Section 3 then goes on to describe and develop the proposed mathematical model for total costs, including purchasing costs, backordering costs and holding costs that are encountered in inventory replenishment by wholesalers and develops the decision rules for trans-shipments. Section 4 shows how these decision rules could be applied in practice and implications for management. Section 5 concludes with a discussion of the effectiveness and limitations of the decision model, with suggestions for further research.

2. Literature review

Thomas and Griffin [25] reviewed the literature addressing coordinated planning between two or more stages of the supply chain, placing particular emphasis on models that would lend themselves to a total supply chain model. Even if a given supply chain network is under a single ownership and centralized control, the integrative view of supply chain management obligates its managers to solve multiple stage inventory problems of a scale for which researchers have yet to identify consistently efficient solution procedures [7,8]. While Köchel et al. [13] stressed the importance of organizing shipments of resource units between the nodes of the logistics network; Weber [26] provided complementarity and substitutability in the shipment related problem. Burton and Banerjee [5] examined the cost effects of two lateral trans-shipment approaches in a two-echelon supply chain network and discovered that a lateral shipment approach was considerably superior to a policy of no such shipments, albeit at the expense of increased transportation activity. Minner and Silver [17] evaluated two simple extreme trans-shipment strategies and developed an analytical approach for estimating the approximate total expected costs. The majority of research papers that deal with goods replenishment issues assume a decision rule is applied when asking for a shipment from other wholesalers [7,14,19].

In organizations, high quality information that is accurate, timely, complete, and consistent and supported by the appropriate analytic tools is critical for managerial decision making [16]. For example, information technology has a significant role to play and appropriate evaluation and decision tools need to be provided for decisions on efficient product development. Taking into account the trend of customer demands and supply of goods for production purpose, Xu et al. [27] proposed a decision model to support product design and indicated the importance of a reliable forecast related to future demands in determining the design criteria. The data repositories, data warehousing and enterprise resource planning implementations improve the potential of decision support capabilities and decision support benefits [11]. In an enterprise value chain that comprises suppliers and customers, integrated information systems are required among the partners in order to deal with the market demand in a responsive way. The importance of a responsive decision model is that it is able to forecast the expected cost of the goods covering inventory and backorder cost [22]. Kimbrough, Wu and Zhong [12] investigated the deployment of artificial agents to manage the flow of goods across the supply chain, minimizing bull-whip effect which can be detrimental to the efficiency in performance of the whole logistics systems. However, it is mainly based on deterministic demand and fixed lead-time and the stochastic phenomenon has been less considered. Reliable information sharing and information quality play an important role in securing performance and achieving total lowest cost in the entire network [15]. Efficient and real-time information sharing can enhance the development of a responsive decision support model and decision models based on mathematical calculations have the potential to support managers in making critical decisions. This paper considered the impact of uncertainty in logistics systems and develops a decision support model for wholesale inventory managers in making lateral trans-shipment decisions in a pragmatically easy method.

Archibald et al. [1] proposed using a stochastic dynamic program to optimize the decision of whether to laterally trans-ship. A simulation with a heuristics approach was evaluated by Tagaras and Cohen [24]. According to Axsäter [2], when a wholesaler cannot supply goods to a retailer, lateral trans-shipment can take place and he proposed a method for optimizing the control policy of inventory replenishment. His model developed decision rules for lateral trans-shipments, aiming to evolve an integrated approach for supporting decisions regarding trans-shipment of goods. The underlying concept of the proposed decision rule by Axsäter ([3]: 1174) is difficult to visualize and adopt practically.

Numerical investigations for small networks show that the rule substantially outperforms the no pooling and complete pooling policies. We have looked at other similar models such as those models considered by Evers [9] and Minner et al. [19] but they examine the case of lost sales. Evers [9] restricts his analysis to two locations only and Minner et al. [19] propose a more flexible heuristic for a multi-location setting, but assume that demand has to be satisfied only just before the new replenishment order arrives. For (R,Q) ordering policies it is often difficult to optimize both ordering parameters and the trans-shipment policy. Also, the method is found to be conservative in its results. Considering an optimal (R,Q) ordering policy under complete pooling Olsson [21] analyzed the steady state nature of the system by assuming that the lead time for an order to arrive is exponentially distributed. However, due to the problem complexities the optimal solution is restricted to systems with only two locations. Chiu and Huang [6] considers a system with more than two echelons but they then restrict focus to a single location of each echelon and assume that unmet demand can be satisfied through trans-shipments if required. Grahovac and Chakravarty [10] propose the base stock level for the replenishments and created the K threshold for triggering trans-shipment requests. When the inventory level of a retailer drops below K, they first try to receive an emergency order with a shorter lead time from the upper echelon, and only use a lateral trans-shipment if this is not possible. However, evaluation of this K threshold approach shows that it is not that reliable and depends largely on whether they are identical and non-identical retailers.

In summary, the extant literature has to some extent dealt with inventory replenishment policies and decisions for wholesaler operations, however, there are limitations that hinder the application of most of them. Consequently, wholesaler inventory managers still use ad hoc methods to make decisions and there is still a need for a practical decision support system to support effective and informed decisions. This paper studies the sourcing decisions of a wholesaler in fulfilling retailer demand and provides a pragmatic approach deriving simple decision rules that are conveniently adoptable by wholesaler inventory management.

3. Model development

When a wholesaler maintains a limited inventory in order to be cost competitive, it constantly needs to externally source to fulfill retailer demand and avoid inventory shortage. We consider a context where there are a finite number of wholesalers operating autonomously in a local region. Following Axsäter [4], we assume that the retailer demand from these wholesalers follows a Compound Poisson distribution. The wholesalers usually replenish their stocks from suppliers, requiring a lead time in order to supply the goods. They may also opt to purchase laterally from other wholesalers in the same cost center which, for the purpose of the model, we assume, due to their close proximity, will provide an instant supply of goods. Such transactions among wholesalers are called lateral trans-shipments and they are more expensive than the usual supplier's prices.

We assume that all wholesalers apply a periodic review policy described by Rosenshine and Obee [23] to replenish from external

suppliers. To formulate the periodic review policy, we review the unsatisfied demands in the previous period, the inventory position, and the expected demand in the current period, and then make the order at the beginning of this period. Unsatisfied demands or surplus orders in this period will be regarded as initial demands or will be added to the inventory position of the next period. If the wholesaler does not employ a (R,Q) review policy, its inventory position could be lower than R of (R,Q) policy or even close to zero, which can decrease the holding costs of the wholesaler. However, the wholesaler that uses a periodic policy has the possibility that its inventory position may be lower than zero. This potentially increases the back-order cost. One possible solution is to make trans-shipment from other wholesalers so as to make the inventory position of the wholesaler non-negative as soon as possible.

We define the following notation used in the model development.

W_i	the i th wholesaler,
N_i	the total number of suppliers to the wholesaler W_i ,
S_{ij}	the j th supplier of the wholesaler W_i ,
p_{ij}	the unit selling price by S_{ij} to W_i ,
q_{ik}	the unit intra-shipment cost for W_i to intraship from W_k ,
b_i	unit back-order cost at W_i per unit time,
h_i	unit holding cost for W_i per unit time,
t_0	start of the scheduling period, $t=0$,
$g_{ij}(t)$	delivery lead time probability mass function of S_{ij} ,
L_{ij}	lead time of S_{ij} with duration equal to L_{ij} times unit time interval,
L_{ij}^{max}	the maximal lead time of S_{ij} ,
$d_i(0)$	the initial retailer demand at $t=0$ appearing at W_i ,
$\lambda_i(t)$	the retailer arrival intensity during the t th time interval at W_i ,
$f_{i,m}^n$	the probability of n retailers arriving at W_i with a total demand of m ,
$\hat{d}_i(t)$	the expected retailer demand at wholesaler W_i in the t th time interval,
$\hat{D}_{ij}$	the expected retailer demand at wholesaler W_i over L_{ij}^{max} .

Following Axsäter [3], we assume retailer demand is a Compound Poisson process and retailers arrive at the wholesaler at an arrival intensity of λ . Assuming the probability of n number of retailers arriving at W_i wholesaler during a time interval of length t is $P_{d_{i,t}}(n)$,

$$P_{d_{i,t}}(n) = \exp^{-\lambda_i t} \frac{\{\lambda_i t\}^n}{n!}. \quad (1)$$

If the conditional probability of n number of retailers requires m demand is given by $P_{d_{i,t}}(m|n) = f_{1,m}^n$ then the probability of retailer demands at the W_i wholesaler during the time interval t is,

$$P_{d_{i,t}}(m) = \sum_{n=1}^m P_{d_{i,t}}(m|n) P_{d_{i,t}}(n) = \sum_{n=1}^m e^{-\lambda_i t} f_{1,m}^n(n) \frac{\{\lambda_i t\}^n}{n!}. \quad (2)$$

The expected retailer demand at W_i wholesaler during a time t is given by

$$\hat{d}_i(t) = \sum_{m=1}^{\infty} m P_{d_{i,t}}(m).$$

Applying the function of $P_{d_{i,t}}(m)$ from Eq. (2),

$$\hat{d}_i(t) = \exp^{-\lambda_i t} \sum_{m=1}^{\infty} \sum_{n=1}^m \frac{\{\lambda_i t\}^n}{n!} m f_{1,m}^n(n). \quad (3)$$

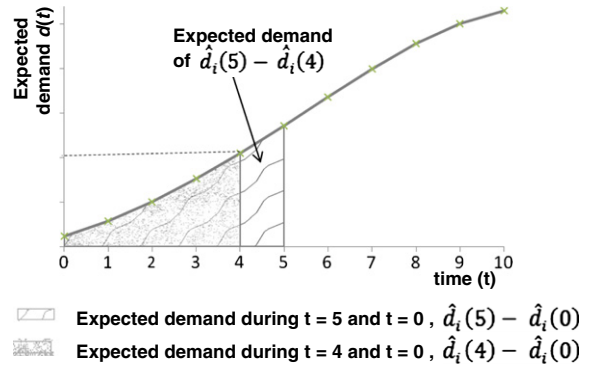


Fig. 1. Diagram of expected demand at the wholesaler W_i .

At $t=0$, demand with the wholesaler W_i $d_i(0)$ is the outstanding demand carried forward from the previous scheduling period and that needs to be urgently fulfilled.

If the supplier, S_{ij} of the wholesaler W_i takes L_{ij} lead time to deliver the order and assuming that the probability mass function of delivery lead time of S_{ij} is $g_{ij}(t)$ then the expected lead time of $E(L_{ij})$ is given by

$$E(L_{ij}) = \int_{t=0}^{L_{ij}} t g_{ij}(t) dt.$$

Let's consider that one scheduling period is the maximum lead time of S_{ij} , L_{ij}^{max} . Then the expected demand at W_i during one scheduling period is

$$\hat{D}_{ij} = \sum_{k=1}^{L_{ij}^{max}} \hat{d}_i(k).$$

From Eq. (3),

$$\hat{D}_{ij} = \sum_{k=1}^{L_{ij}^{max}} e^{-\lambda_i k} \sum_{m=1}^{\infty} \sum_{n=1}^m \frac{\{\lambda_i k\}^n}{n!} m f_{1,m}^n(n). \quad (4)$$

Fig. 1 presents a schematic diagram of the expected retailer demand at wholesaler W_i and Table 1 presents the expected demand for each time period. The calculations in the following sections refer back to these for better understanding.

3.1. Costs of supply

Inventory costs of a wholesaler consist of three components: purchasing costs of P_{ijk} of the orders to suppliers and intra-shipments from the other wholesalers, backordering costs of B_{ijk} for unfulfilled retailer demands and holding costs of H_{ijk} for carrying inventory for potential demands.

$$C_{ijk}(x) = P_{ijk}(x) + B_{ijk}(x) + H_{ijk}(x) \quad (5)$$

Table 1
Expected demand at the wholesaler W_i .

Time period	Expected demand
$[t_0 - t_1]$	$\hat{d}_i(1)$
$[t_1 - t_2]$	$\hat{d}_i(2) - \hat{d}_i(1)$
$[t_2 - t_3]$	$\hat{d}_i(3) - \hat{d}_i(2)$
$[(L_{ij}^{max} - 2) - (L_{ij}^{max} - 3)]$	$\hat{d}_i(L_{ij}^{max} - 1) - \hat{d}_i(L_{ij}^{max} - 2)$
$[(L_{ij}^{max} - 1) - (L_{ij}^{max})]$	$\hat{d}_i(L_{ij}^{max}) - \hat{d}_i(L_{ij}^{max} - 1)$

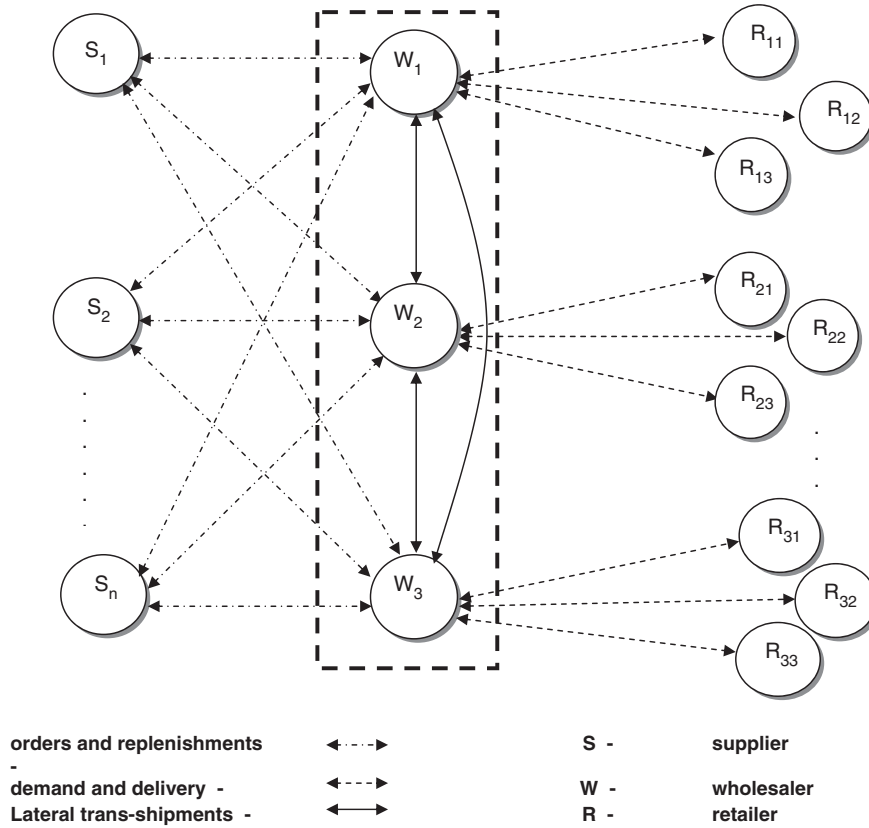


Fig. 2. A schematic diagram of the supply chain of wholesalers, retailers and suppliers.

We consider that the maximum lead time taken by the supplier is the scheduling period for the wholesaler's decision. In our model, the period to update the wholesaler is set to the maximal lead time of the given supplier. So all the orders made should be received in the period. W_i therefore there will be no outstanding orders from its suppliers at the initial point of a new period, which will simplify the total cost model and decision rules for intra-shipment. So the inventory level of the wholesaler is equal to its inventory position.

The context of the wholesale operation consists of multiple retailers, wholesalers and suppliers as shown in Fig. 2. All the wholesalers belong to the same cost center. The dashed black lines represent possible intra-shippments within the cost center. For simplicity of this decision model, we assume that when the wholesaler, W_i makes intra-shippments, this involves a selected single another wholesaler. Similarly, W_i places orders with a single supplier instead of using multiple suppliers.

3.1.1. Purchasing costs

Purchasing costs are incurred when the wholesaler does not have sufficient stock to fulfill the retailer demand and place the orders for the unfulfilled demand from another supplier or another wholesaler. Hence, the purchasing cost has two components depending on the source of supply: supplier or another wholesaler. If the unit cost of purchasing from the supplier S_j is p_{ij} , the unit cost of trans-shipping from the wholesaler W_k is q_{ik} and the trans-shipped quantity is x units then the purchasing cost of $P_{ijk}(x)$ is given by

$$P_{ijk}(x) = p_{ij}(d_i(0) - l_i(0) + \hat{D}_{ij} - x) + q_{ik}(x) \quad (6)$$

where $d_i(0)$ and $l_i(0)$ are the outstanding order quantity and the inventory level of the W_i at $t=0$.

3.1.2. Backordering costs

The backordering cost is incurred when the excess retailer orders that cannot be fulfilled immediately are held on the books until the next shipment arrives [20]. As Axsäter [3], we assume that excessive retailer demands that cannot be fulfilled by the stock in hand and transshipments (if decided,) are backordered. For the initial back-ordered quantity of $(d_i(0) - l_i(0) - x)$ the back-ordering time is the expected arrival time of order from the supplier. We assume that intra-shipping cost from another wholesaler is significantly higher than the purchasing cost from suppliers. Hence, only the urgent orders are intra-shipped and others are sourced from suppliers. We also assume that intra-shipped orders have zero lead time. Hence, at the beginning of the scheduling period, any outstanding order that cannot be fulfilled by the local inventory and intra-shipped orders, generate backordering costs, until the expected supplier order supply entry day. Subsequent retailer demands expected to arrive during the time range of $0 \leq t \leq E(L_{ij}) - 1$ are not fulfilled due to stock-out status and consequently they generate back-ordering cost depending on the back-order time.

Based on the expected retailer demand as shown in Fig. 1 and Table 1, the following Table 2 presents the back-order time for the unfulfilled demand at the wholesaler, W_i .

If B_{ijk} is the backorder cost for the wholesaler W_i for the orders placed with the supplier S_j after the intrashippments from the W_k and b_{ij} is the unit back-order cost then,

$$B_{ijk}(x) = b_{ij}(d_i(0) - l_i(0) - x)E(L_{ij}) + b_{ij} \int_{t=1}^{E(L_{ij})-1} \{d(t) - \hat{d}(t-1)\} \{E(L_{ij}) - t\} dt. \quad (7)$$

As shown in Eq. (7), the two components of the backordering cost are the back-order cost at $t=0$ and the backorder cost for the expected demand until one period before the expected lead time of the supplier. The expected demand at the period of expected lead

Table 2

Backorder time period for the unfulfilled demand.

Time period of demand	Demand	Backorder period (e.g. number of days)
at t_0	$(d_i(0) - l_i(0) - x)$	$E(L_{ij})$
$[t_0 - t_1]$	$\hat{d}_i(1)$	$E(L_{ij}) - 1$
$[t_1 - t_2]$	$\hat{d}_i(2) - \hat{d}_i(1)$	$E(L_{ij}) - 2$
$[t_2 - t_3]$	$\hat{d}_i(3) - \hat{d}_i(2)$	$E(L_{ij}) - 3$
$\{[E(L_{ij}) - 2] - [E(L_{ij}) - 3]\}$	$\hat{d}_i(E(L_{ij}) - 2) - \hat{d}_i(E(L_{ij}) - 3)$	2
$\{[E(L_{ij}) - 1] - [E(L_{ij}) - 2]\}$	$\hat{d}_i(E(L_{ij}) - 1) - \hat{d}_i(E(L_{ij}) - 2)$	1

time can be fulfilled with the received supply from the supplier. Any retailer demands arriving after the expected lead time of supplier do not incur any backordering cost but will generate holding costs that will be discussed separately below.

3.1.3. Holding costs

Holding cost is the inventory carrying cost which consists of the costs of providing physical space to store items, taxes and insurance, breakage and other damages and opportunity cost of alternative investment [20]. During the scheduling period, holding costs are incurred after the order is received at the expected lead time of the supplier and the outstanding demands are delivered i.e. during the time period of $E(L_{ij}) + 1 \leq t \leq L_{ij}^{\max}$. Based on the expected demand as shown in Fig. 1 and Table 1, the following Table 3 presents the holding cost for the stock received from the supplier at $E(L_{ij})$.

If the total holding cost of H_{ijk} during the scheduling period and the unit holding cost of the W_i wholesaler is h_i then,

$$H_{ijk}(x) = \int_{t=E(L_{ij})+1}^{L_{ij}^{\max}} \left\{ \hat{d}(t) - \hat{d}(t-1) \right\} \{t - E(L_{ij})\} h_i dt \quad (8)$$

3.2. Decision rule for intra-shipment orders

By substituting the functions of purchasing costs, backordering costs and holding costs in Eq. (5), the total cost during the scheduling period is given by,

$$\begin{aligned} C_{ijk}(x) &= p_{ij} (d_i(0) - l_i(0) + \hat{D}_{ij} - x) + q_{ik}(x) \\ &\quad + (d_i(0) - l_i(0) - x) E(L_{ij}) b_i + b_i \int_{t=0}^{E(L_{ij})-1} \hat{d}(t) - \hat{d}(t-1) \{E(L_{ij}) - t\} dt \\ &\quad + \int_{t=E(L_{ij})+1}^{L_{ij}^{\max}} \left\{ \hat{d}(t) - \hat{d}(t-1) \right\} \{t - E(L_{ij})\} h_i dt \\ C_{ijk}(x) &= \left\{ -p_{ij} - E(L_{ij}) b_{ij} + q_{ik} \right\} x \\ &\quad + p_{ij} (d_i(0) - l_i(0) + \hat{D}_{ij}) + \{d_i(0) - l_i(0)\} E(L_{ij}) b_i \\ &\quad + b_i \int_{t=1}^{E(L_{ij})-1} \left\{ \hat{d}(t) - \hat{d}(t-1) \right\} \{E(L_{ij}) - t\} dt \\ &\quad + \int_{t=E(L_{ij})+1}^{L_{ij}^{\max}} \left\{ \hat{d}(t) - \hat{d}(t-1) \right\} \{t - E(L_{ij})\} h_i dt. \end{aligned} \quad (9)$$

Table 3

Holding period of the stock received from the supplier.

Time period of demand	Demand	Holding period (e.g. in number of days)
$\{E(L_{ij}) + 1\} - E(L_{ij})$	$\hat{d}_i(E(L_{ij}) + 1) - \hat{d}_i(E(L_{ij}))$	1
$\{E(L_{ij}) + 2\} - \{E(L_{ij}) + 1\}$	$\hat{d}_i(E(L_{ij}) + 2) - \hat{d}_i(E(L_{ij}) + 1)$	2
$\{E(L_{ij}) + 3\} - \{E(L_{ij}) + 2\}$	$\hat{d}_i(E(L_{ij}) + 3) - \hat{d}_i(E(L_{ij}) + 2)$	3
$\{L_{ij}^{\max} - 1\} - \{L_{ij}^{\max} - 2\}$	$\hat{d}_i(L_{ij}^{\max} - 1) - \hat{d}_i(L_{ij}^{\max} - 2)$	$E(L_{ij}^{\max} - 1) - E(L_{ij})$
$L_{ij}^{\max} - \{L_{ij}^{\max} - 1\}$	$\hat{d}_i(L_{ij}^{\max}) - \hat{d}_i(L_{ij}^{\max} - 1)$	$E(L_{ij}^{\max}) - E(L_{ij})$

With respect to the quantity of trans-shipment (x), the total cost function as shown in Eq. (9) is linear. When the tangent of the above linear function $(-p_{ij} - E(L_{ij}) b_i + q_{ik})$ is negative the total cost decreases with the increasing number of trans-shippments.

Hence the decision rule for intra-shippments is

$$\begin{aligned} -p_{ij} - E(L_{ij}) b_i + q_{ik} &< 0 \\ q_{ik} &< p_{ij} + E(L_{ij}) b_i. \end{aligned} \quad (10)$$

In summary, if the above decision rule shown in Eq. (10) is satisfied, the higher the quantity of trans-shipment the lower the total cost for the wholesaler W_i . In other words, the wholesaler W_i should decide to fulfill the demand by ordering only from the supplier S_{ij} if the above decision rule is not satisfied.

3.3. Decision rules for the favorite wholesaler and the optimum size of trans-shipment

According to the total cost function shown in Eq. (9), if the wholesaler W_i satisfied the decision rule in Eq. (10), the higher the trans-shipment quantity, the lower the total cost to the wholesaler W_i . Furthermore, it suggests the preference for the lowest unit cost of trans-shipment, q_{ik} and in turn trans-ship from the wholesaler W_k who provides at the lowest cost.

We only order the initial outstanding urgent demand at $t=0$ which is $d_i(0)$ from another wholesaler due to the increased unit cost of trans-shipment from another wholesaler compared with the unit purchasing cost from suppliers. Even though the maximum benefits are received when the highest quantity is trans-shipped, this maximum quantity of trans-shipment has a practical limitation – the available stock with the wholesaler W_k . For simplicity of the model, we assume that the preferred other wholesaler, has sufficient stock to deliver the trans-shipment. Hence the limiting parameter of the trans-shipment order is the outstanding demand at $t=0$. The optimum size of the trans-shipment, μ_k is given by

$$\mu_k = d_i(0) - l_i(0). \quad (11)$$

Therefore, in this model the size of the trans-shipment can be either 0 or μ_k . When the decision rule in Eq. (10) is not satisfied there will not be any trans-shipment and the maximum of μ_k when it is satisfied.

3.4. Decision rule for the selection of the supplier

According to Fig. 1, the wholesaler W_i is able to purchase from any one of its suppliers. The selection decision is derived by the global minimization of the total cost function in Eq. (9) with a fixed x ($x \neq 0$) and a known W_k .

Hence, the decision rule is given by the condition that satisfies

$$C_i = \min(C_{i1}, C_{i2}, \dots, C_{ij}, \dots, C_{iN_i}) \quad (12)$$

Where N_i is the number of suppliers of the wholesaler W_i .

4. Scenario analysis and discussion

In order to illustrate the pragmatic use of the decision rules derived above, let's consider the decision making process of the wholesaler XYZ. Due to intense competition in the market, accessing confidential business information such as order and procurement data is extremely difficult to obtain. We consider therefore, a hypothetical wholesaler of XYZ and assume it has a constant retailer arrival intensity so $\lambda(t) = \lambda$ for any t . The probability of n number of retailers arriving at the wholesaler XYZ with demand m , $f_{1,m}^n$ can be

calculated mathematically by using the method proposed by Axsäter [4]: 80–83). According to Axsäter [4], the probability distribution of demand, $D(t)$ can be shown by a negative binomial distribution as shown below.

$$P(D(t) = k) = \frac{r(r+1)\dots(r+k-1)}{k!} (1-p)^r p^k,$$

where r can be any positive number and $0 < p < 1$.

Alternatively, the probabilities can also be calculated by using historical retailer demand records of XYZ from their inventory management. In addition, using the historical retailer orders, sales, procurement and inventory accounting records of XYZ, the values of retailer arrival intensity (λ), unit holding cost (h_1) and the back-ordering cost (b_1) can be estimated. Let's take $\lambda = 3$, $h_1 = 2$, and $b_1 = 2$ for the wholesaler XYZ. Since the retailer demand is essentially finite we set the upper bound of the retailer demand at the wholesaler XYZ as $d_1^{max} = 30$.

4.1. Case 1 – wholesaler XYZ operating with one supplier and one other wholesaler

First, we consider a somewhat limited context where wholesaler XYZ having one other wholesaler W_2 and one supplier S_1 , let's assume the maximum lead time of $L_1^{max} = 10$, unit purchasing cost from the supplier 1, p_{11} is 2.2 and the inventory level of XYZ at $t = 0$, $I_1(0) = 0$ and the sudden demand at $t = 0$, $d_1(0) = 6$.

Table 4 presents the decision making process of the wholesaler XYZ for different values of the unit intra-shipment cost from the other wholesalers, q_{1k} and the expected lead time from the supplier 1, $E(L_{11})$.

Fig. 3 and Table 5 show the cost function for those different values of q_{1k} and $E(L_{11})$ and indicate that only when the decision rule applies the total cost decreases with the increasing size of trans-shipment from the other wholesaler. The total cost functions of C_{144} and C_{122} demonstrate a scenario in which trans-shippments should not be made by the wholesaler XYZ and the other C_{111} and C_{133} satisfy the decision rule to choose to trans-ship from another wholesaler.

4.2. Case 2 – wholesaler, XYZ operating with one supplier and multiple other wholesalers

When the decision for trans-shippments is made, the immediate following decision that needs to be made is with which other wholesaler to place the intra-shipment order. Considering a scenario where the wholesaler, XYZ operates in a system with one chosen supplier and multiple other wholesalers, we can demonstrate the application of the decision for the favorite other wholesaler following the rule in Eq. (12).

Table 6 shows calculated values for the total cost and Fig. 4 demonstrates how this decision rule, derived earlier, works. In a context with one supplier, S_1 and four other wholesalers, W_1 , W_2 , W_3 , the graph shows the total cost functions for varying values for q_{ijk} for 4 different other wholesalers.

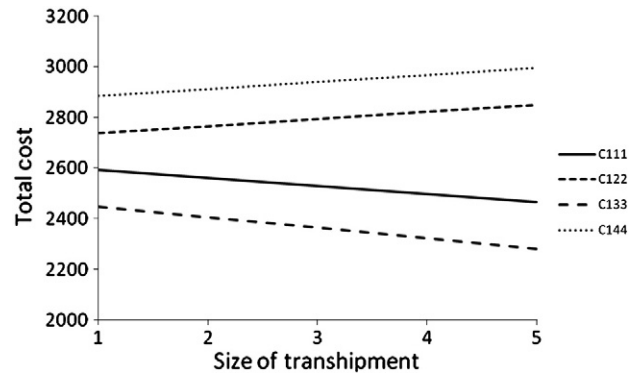


Fig. 3. Total cost and trans-shipped units for different q_{ik} and $E(L_{ij})$.

The total cost function of C_{114} is increasing with the size of trans-shippments thus omitted because that does not satisfy the primary decision rule. From the others, C_{114} at the bottom gives the minimum cost for any amount of trans-shipped units which makes W_4 the favorite wholesaler for XYZ in alignment with the corollary decision presented in Eq. (12).

4.3. Case 3 – wholesaler, XYZ operating with multiple suppliers and multiple other wholesalers

Considering a more general scenario of wholesaler XYZ operating in a system with multiple wholesalers and suppliers, the decision for the favorite supplier and wholesaler that provides the minimum total cost is critical. Such decisions are more complicated in an environment as shown in Fig. 2. Fig. 5 plots the total cost function for different values of q_{ijk} and p_{ij} representing different wholesalers and suppliers and Table 7 presents the corresponding values. Since the total cost functions of C_{115} , C_{125} and C_{135} increase with the size of trans-shipment, they are excluded from further consideration. The remaining cost functions satisfy the primary decision rule for trans-shipment by XYZ. If the outstanding demand at $t = 0$ is 6, C_{111} at the bottom provides the lowest cost for the quantity of 6 units of trans-shipment. Therefore, XYZ should trans-ship 6 units from the other wholesaler 1 and backorder D_{max} of 30 units from the supplier 1.

This approach has implications for managers in logistics operations. The pragmatic adoptability is important in development of theoretical models. For inventory management, the decisions on lateral trans-shippments are needed to be made systematically without following ad hoc methods and the above three case scenarios represent how the proposed model may be applied in different real situations in logistics operations. The less complexity of data requirements and mathematical calculations and the visualization of the underlying concepts of this approach through the graphical plots show the pragmatic benefits of this approach in real contexts. In particular, the calculations can be done using commonly used spreadsheet software such as MS Excel and Fig. 6 presents a screenshot of how MS Excel is used to make the decision for the complex

Table 4

Decision rule for the wholesaler XYZ in a context with one other wholesaler and one supplier.

	$q_{11} = 5, E(L_{11}) = 3$	$q_{12} = 9, E(L_{11}) = 2$	$q_{13} = 6, E(L_{11}) = 4$	$q_{14} = 7, E(L_{11}) = 2$
Total cost function	C_{111}	C_{122}	C_{133}	C_{144}
Testing the decision rule	$q_{12} < p_{11} + E(L_{11})b_1$ decision rule satisfied	$q_{12} > p_{11} + E(L_{11})b_1$ decision rule dissatisfied	$q_{12} < p_{11} + E(L_{11})b_1$ decision rule satisfied	$q_{12} > p_{11} + E(L_{11})b_1$ decision rule dissatisfied
Decision	Maximum quantity of 6 units to be trans-shipped from the wholesaler W_2	No trans-shippments. All ordered from the supplier S_1	Maximum quantity of 6 units to be trans-shipped from the wholesaler W_2	No trans-shippments. All ordered from the supplier S_1

Table 5

Total cost for different trans-shipment unit cost and expected lead time of the supplier.

Size of trans-shipment, x	0	10	20	30	40
C_{111} for $q_{11} = 5, E(L_{11}) = 3$	2592.88	2560.88	2528.88	2496.88	2464.88
C_{122} for $q_{12} = 9, E(L_{11}) = 2$	2737.82	2765.82	2793.82	2821.82	2849.82
C_{133} for $q_{13} = 6, E(L_{11}) = 4$	2447.94	2405.94	2363.94	2321.94	2279.94
C_{144} for $q_{14} = 7, E(L_{11}) = 2$	2882.76	2910.76	2938.76	2966.76	2994.76

Table 6

Calculated values for total cost for XYZ having one supplier and multiple other wholesalers.

Size of trans-shipment, x	0	2	4	6	8
C_{111}	2592.88	2590.48	2588.08	2585.68	2583.28
C_{112}	2592.88	2587.48	2582.08	2576.68	2571.28
C_{113}	2592.88	2588.48	2584.08	2579.68	2575.28
C_{114}	2592.88	2589.48	2586.08	2582.68	2579.28

scenario of XYZ operating in a context of multiple suppliers and wholesalers.

5. Conclusion

The decision support system developed in this article serves the inventory management of wholesaler operations in making decisions on whether to trans-ship outstanding urgent retailer demands or back order from suppliers in full. The main advantage of this decision support system is the ease of application by wholesaler inventory management. The decision is driven by the important cost minimization objectives, the simplicity of the rules and the need for less cumbersome data inputs to the model, underpin the ease of adoption. The main decision rule needs only the unit purchasing cost from suppliers, unit trans-shipment cost from another wholesaler, own unit backordering cost, and the expected lead time from its suppliers. The corollary decisions on deciding the favorite other wholesaler and supplier, derived in this article can also be easily adopted by inventory management without performing complex mathematical calculations. Moreover, common spreadsheet software such as MS Excel can be used for these calculations as shown above.

We are aware of the limitations of this study. First, we approximated that the probability distribution of retailer arrival at the wholesaler has a Compound Poisson distribution. Second, the suggested method of estimating backorder costs and holding costs was based

on previous cost records. We assumed that the estimates based on old records would hold true during the forecasting period but the current dynamic business environment challenges that assumption. Furthermore, we assumed that the selected other wholesaler is able to supply the trans-shipment order in full thus no partial orders with multiple wholesalers were considered. However, wholesalers can place partial orders of trans-shipments with multiple other wholesalers for reasons of risk mitigation. Future research has the opportunity to accommodate the situation of partial orders with multiple suppliers and other wholesalers and to further develop the proposed decision support system in the present study.

There is potential for developing this model into a software program for wide adoption by logistics organizations. In a standalone program, the user interface should allow the input fields for organization-specific dynamic parameters such as previous demand records, unit backordering, holding and purchasing costs and supplier lead times etc. In a wider scale, the proposed decision rule algorithm can be integrated as a module of the existing enterprise resource planning applications with the capability of extracting data from previous transaction records in the internal databases, thereby extending the capability of achieving more reliable and accurate expected costs in inventory management operations. Such extended software applications should be scalable to handle different scenarios and upgradable to accommodate the intended potential future developments of this approach covering partial order situations.

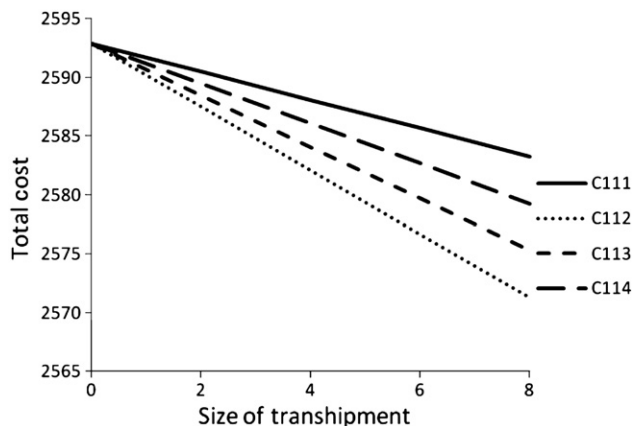


Fig. 4. Total cost functions — XYZ operating in a system with one supplier and multiple other wholesalers.

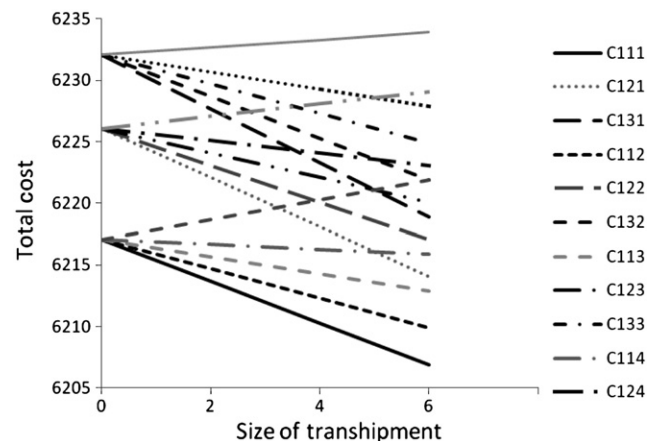
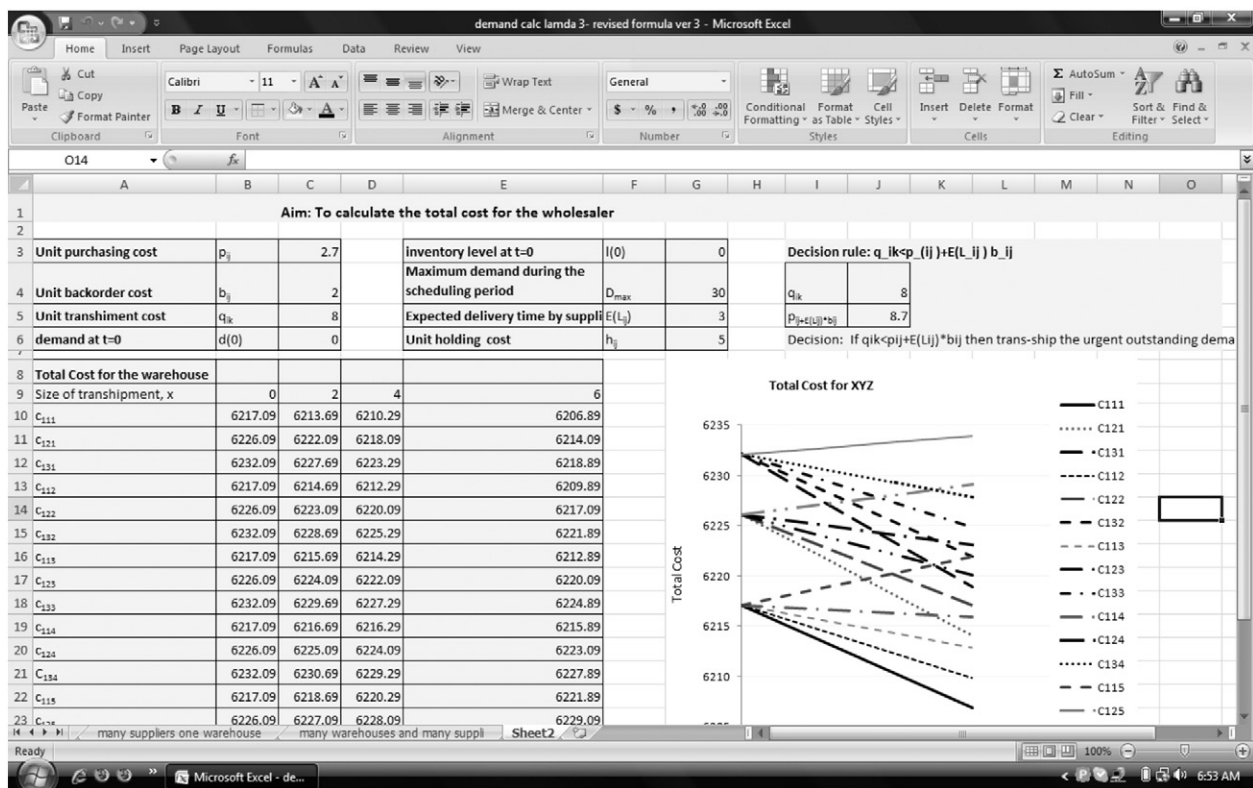


Fig. 5. Total cost functions for XYZ in a system with multiple suppliers and multiple wholesalers.

Table 7

Calculated values of total cost for the wholesaler for different options in a multi supplier and multiple wholesaler system.

Size of trans-shipment, x	0	2	4	6
C_{111}	6217.09	6213.69	6210.29	6206.89
C_{121}	6226.09	6222.09	6218.09	6214.09
C_{131}	6232.09	6227.69	6223.29	6218.89
C_{112}	6217.09	6214.69	6212.29	6209.89
C_{122}	6226.09	6223.09	6220.09	6217.09
C_{132}	6232.09	6228.69	6225.29	6221.89
C_{113}	6217.09	6215.69	6214.29	6212.89
C_{123}	6226.09	6224.09	6222.09	6220.09
C_{133}	6232.09	6229.69	6227.29	6224.89
C_{114}	6217.09	6216.69	6216.29	6215.89
C_{124}	6226.09	6225.09	6224.09	6223.09
C_{134}	6232.09	6230.69	6229.29	6227.89
C_{115}	6217.09	6218.69	6220.29	6221.89
C_{125}	6226.09	6227.09	6228.09	6229.09
C_{135}	6232.09	6232.69	6233.29	6233.89

**Fig. 6.** MS Excel spreadsheet for calculating the favorite supplier in case 3.

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